

May 14, 2024

Comments of the Vehicle-Grid Integration Council on ComEd BE Pilots

I. INTRODUCTION.

The Vehicle-Grid Integration Council (VGIC)¹ is a 501(c)(6) nonprofit member-based association focused on accelerating the role of smart electric vehicle (EV) charging and discharging (i.e., vehicle-grid integration or “VGI”) through policy development, education, outreach, and research. Scaling VGI is an essential part of transportation electrification and supports the following key goals:

- **Benefit EV drivers and fleet owners** by reducing the total cost of ownership and providing additional value streams like backup power.
- **Decarbonize the transportation sector** by accelerating EV adoption through additional benefits to EV drivers and fleet owners.
- **Support the evolving electric grid** by providing essential grid services as variable energy generation and distributed energy resource penetration increases.
- **Increase affordability** by right-sizing infrastructure investments and placing downward pressure on rates, in turn reducing electricity bills for all customers.
- **Improve community resiliency and security** by utilizing EVs to provide emergency power during planned and unplanned grid outages.
- **Foster economic activity** by catalyzing innovation, competition, and market transformation.

With the proper utility and regulatory support and coordination, these goals can be achieved, and individual EV drivers and EV fleets in Illinois can play a critical role in the acceleration of both transportation electrification and grid decarbonization. **Our vision for VGI encompasses the following key elements:**

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., BorgWarner, Fermata Energy, Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, Customized Energy Solutions, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, FreeWire Technologies, Inc., GridWiz, Kaluza, Landis + Gyr, Leapfrog Power, Inc., Nuvve Holding Corporation, Peak Power, PowerFlex, Qcells, Stellantis N.V., Sumitomo Electric, The Mobility House, Toyota Motor North America, Inc., WeaveGrid, Hoosier Energy, Peninsula Clean Energy, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>)

- **Ensure customer mobility needs are satisfied:** Drivers and fleets can support the grid without compromising mobility needs.
- **Managed charging will benefit EV drivers and fleet operators:** Drivers and fleets in Illinois will be provided a menu of opportunities to align charging with the times of day when electricity prices are low, reducing EV and fleet operating costs by as much as 50% compared to unmanaged charging. Lowering the total ownership cost of EV ownership will accelerate overall EV adoption by drivers and fleet managers in Illinois. Furthermore, managed charging reduces the need for expensive distribution system upgrades.
- **EVs provide reliable emergency power during blackouts:** During extreme weather blackouts or other power outages, EV drivers and fleets can utilize bidirectional charging capabilities to send energy to a home, building, or microgrid, serving as a generator and providing safe emissions-free backup power for households and communities.
- **VGI strategies right-size charging infrastructure investments and accelerates deployment:** Smarter management of EV charging will help maximize the impact of utility infrastructure investments, which encourages wider access to EV charging.
- **VGI enables EVs to provide valuable services to the grid and generate revenue:** V2X (vehicle-to-everything bidirectional charging) enables EVs to both charge and feed power back to the grid, providing grid services such as demand response, peak shaving, frequency regulation, and more. A number of utilities across the US have implemented programs that provide compensation for these valuable grid services.

VGIC commends ComEd for hosting the BE Plan 2 Stakeholder Collaboration Sessions and appreciates the opportunity to provide feedback on ComEd’s proposed BE Plan pilots.² VGIC offers the recommendations below to support ComEd as it executes the proposed pilots and continues developing BE Plan 2.

II. DISCUSSION

a. VGIC supports ComEd’s proposed pilots and advises against relitigating the findings in Final Order 22-0432.

On March 23, 2023, the Illinois Commerce Commission (“Commission”) issued Final Order 22-0432 in ComEd’s Beneficial Electrification Plan (“BE Plan”), requiring ComEd to

² ComEd BE Plan 2 Stakeholder Collaboration Session 1. *Beneficial Electrification (BE) Plan 2: Stakeholder Collaboration – Session 1*, April 10, 2024; ComEd BE Plan 2 Stakeholder Collaboration Session 2. *Beneficial Electrification (BE) Plan 2: Stakeholder Collaboration – Session 2*, April 17, 2024.

implement eight pilots (“BE Pilots”) as part of their BE Plan.³ As such, ComEd filed its BE Plan Compliance Filing in May 2023, which outlined the eight planned BE Pilots developed based on stakeholder ideas and Commission directives. ComEd’s BE Pilots included the following eight pilots: (1) School Bus Vehicle-to-Grid, (2) Residential Optimized Charging, (3) Curbside Charging, (4) Ridesharing (or other pilot program aimed at providing EV access to low-income and EJ/R3 customers for whom car ownership is not an option); (5) Backup Power Capabilities, (6) EV EMS, (7) Submetering, and (8) Air Quality Mapping Pilot.⁴

VGIC commends ComEd’s efforts in developing the BE Pilots, which represent important steps forward for Illinois’ ability to leverage VGI technologies. Despite the Commission’s prior approval and explicit direction for the eight BE Pilots and ComEd’s attempted compliance with this directive, participants in the recent BE Plan 2 stakeholder workshops questioned the effectiveness of ComEd’s proposed pilot plans. While VGIC believes it is unfortunate that these pilots have not yet launched in the 14 months since the Final Order, **VGIC recommends ComEd continue its efforts to establish all eight BE Pilots as directed by the Commission’s Final Order.** The BE Pilots should not be subject to cancellation or significant alteration, as that would undermine the Commission’s intent and the state’s clean energy goals.

Moreover, as detailed below, these pilots will yield important lessons that will inform the scale-up to mass-market, territory-wide customer offerings that will play a central role in ComEd’s grid of the future. ComEd has an important opportunity to lay a foundation for widespread success in BE Plan 2 and beyond by continuing with the proposed BE Pilots.

b. ComEd should execute and assess the BE Pilots through an intentional lens of scalability.

VGIC reiterates that ComEd should move forward with proposed pilots and not cancel or reduce the scale of any proposed pilots; however, we recommend the following improvement to pilot execution. While findings related to technology demonstration are generally valuable, the implementation and assessment of the BE Pilots must include a clear roadmap from small-scale pilots to large-scale programs. Specifically, **VGIC recommends ComEd develop a Roadmap to Scale for each pilot that includes the following key elements:**

³ Illinois Commerce Commission Final Order 22-0432, *Petition for Approval of Beneficial Electrification Plan under the Electric Vehicle Act, 20 ILCS627/45 and New EV Charging Delivery Classes under the Public Utilities Act, Article IX*. March 23, 2023. See: <https://www.icc.illinois.gov/docket/P2022-0432/documents/349478/files/610872.pdf>

⁴ ComEd Beneficial Electrification Plan Compliance Filing, Pg. 42.
See: <https://www.icc.illinois.gov/docket/P2022-0432/documents/338224/files/589765.pdf>

- Identification of customer engagement, marketing, education, and outreach strategies tested in the BE Pilots that should be preserved in a scale-up to mass-market program.
- Needs assessment for large-scale integrations with third-party technology providers and viable partnership structures.
- Results of customer incentive tests (e.g., monthly vs yearly payments) and identifying the share of upfront vs performance-based incentives that yield the desired customer behavior and/or automated technology response.
- Identification of internal utility operational and process improvements needed to achieve mass-market implementation.

Example learnings that can inform an on-ramp from small-scale pilot to large-scale program include integrating flexible service connection (i.e., “EV EMS”) into the new load request process for sites with exceptionally long energization timelines, streamlining the interconnection process for vehicle-to-grid (“V2G”) charging equipment, improving customer interconnection portals and forms, rate design experiments for vehicle-to-everything (“V2X”) bidirectional charging use cases, and customer access to make-ready funding for sites where BE Pilot technologies are used.

VGIC notes similar pilots across the US have successfully evolved into larger-scale initiatives. For example, in California, Pacific Gas & Electric’s (“PG&E”) tested and demonstrated the capabilities of bidirectional charging technology from various manufacturers and third-party service providers in its V2X Pilots – Phase 1, as is highlighted in PG&E’s most recent VGI annual report.⁵ PG&E is now slated to launch a larger-scale “V2X Pilots – Phase 2,” which aims to enroll more customers and experiment with additional incentive mechanisms. Furthermore, V2G school bus pilots supported by San Diego Gas & Electric (“SDG&E”) later evolved into commercial deployments enrolled in California’s large-scale voluntary emergency demand response program, the Emergency Load Reduction Program.⁶ Lastly, Consolidated Edison’s residential managed charging program, Smart Charge NY, began as a small-scale pilot and has now evolved into a mass-market program available to all residential EV customers.⁷

⁵ PG&E, *Vehicle Grid Integration Report by Pacific Gas and Electric Company (U 39 E)*. March 15, 2023. See: <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M503/K589/503589008.PDF>

⁶ Nuvve Corporation. Blue Bird Delivers North America’s First-Ever Commercial Application of Vehicle-to-Grid Technology in Electric School Bus Partnership with Nuvve and Illinois School Districts. March 23, 2021. <https://nuvve.com/blue-bird-v2g-electric-bus-with-nuvve-and-illinois-school-districts/>

⁷ <https://www.coned.com/en/save-money/rebates-incentives-tax-credits/rebates-incentives-tax-credits-for-residential-customers/electric-vehicle-rewards>

With this in mind, **VGIC recommends that ComEd not only proceed with the proposed BE Pilots, but also adopt an explicit focus on charting a path toward large-scale programs, including identifying necessary steps to support (1) customer engagement, (2) third-party provider integration, (3) incentive and compensation design, and (4) operational enhancements.** While adequate demonstration of VGI technology capabilities is important, ComEd must also pilot programmatic design elements and implementation strategies that can effectively and efficiently evolve into mass-market program offerings in the near future.

c. The Backup Power Capabilities Pilot should proceed but be enhanced to support customers and match the current stage of market development for backup power solutions.

ComEd proposed the eight pilots discussed above in its compliance filing “to unlock the full potential of beneficial electrification.”⁸ VGIC agrees that ComEd will be unlocking significant potential from beneficial electrification, including transportation electrification, via the BE Pilots. As workshop participants noted, ComEd’s approach must coincide with current EV technological advancements, including market developments since the pilots were litigated in 2022 and 2023. Specifically, workshop participants inquired about the need for a Backup Power Capabilities pilot given the commercially-availability of vehicle-to-building (“V2B”) backup power solutions. While VGIC maintains that none of the eight Commission-directed BE Pilots should be canceled or significantly altered, we do recognize that the market for these V2B backup power solutions has evolved considerably since 2022, and with it, the pilot may also need to evolve. With this in mind, rather than limiting the Backup Power Capabilities pilot to “study mobility requirements, bidirectional EV charging stations, and the performance of EVs as backup power sources,”⁹ **VGIC recommends ComEd prioritize (1) customer marketing, education, and outreach for V2B backup power solutions and (2) upfront incentive mechanisms to support equipment adoption of V2B backup power solutions.** This would allow ComEd to execute on the Commission-approved pilot while ensuring its efforts appropriately match the current stage of V2B market development.

III. CONCLUSION.

VGIC appreciates the opportunity to submit these comments on ComEd’s proposed BE Pilots. We look forward to further collaboration with ComEd and other stakeholders in Illinois on this important initiative.

⁸ Ibid.

⁹ Ibid. Pg. 43.



Respectfully submitted,

Zach Woogen
/s/ Zach Woogen
Interim Executive Director
Vehicle-Grid Integration Council (VGIC)
vgicregulatory@vgicouncil.org
(510) 665-7811
www.vgicouncil.org